

Puzzles to Puzzle You (Remastered)

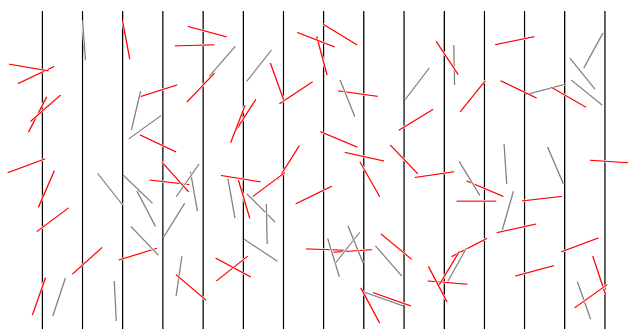
Welcome to the fifth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Buffon's Needle Problem) and pose the next one.

Puzzle #4: Buffon's Needle Problem

Suppose we have a floor made of parallel strips of wood, each the same width w apart. Suppose further we drop a needle of length l onto the floor, where $l \leq w$.

1. What is the probability that the needle will intersect a line between two strips?

$$\frac{2l}{\pi w}$$



Without any circles, we have somehow produced π ! How is that possible?! The underlying reason is that the probability distribution function (PDF) for the needle's orientation is *rotationally symmetric*.

In virtue of this, Buffon's Needle Problem has been used to design Monte Carlo methods for approximating π . Here's [one](#) from the University of Illinois Urbana-Champaign, where you can also read a solution to the problem.

Shout Out My friend Dimitri — a multi-talented engineer, chef, and taste-maker — solved the puzzle while on vacation in Costa Rica. I cannot think of a more perfect way to spend one's evening, lounging on the beach puzzling over geometric probability. To travel — both physically and by one's own imagination — is the purest form of leisure, free from bourgeois coöptation.

Dimitri Leggas
to me

Wed, May 7, 2:38 PM (2 days ago)

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The probability the needle intersects a line is independent of the strip it falls in and vertical position. Then we only need to consider the horizontal position of one point on the needle and angle of the needle.

I assume wlog that the left endpoint of the needle lies in strip. The needle can only possibly intersect at right if the left endpoint is at most l from the right. We dropped uniformly at random so we have probability l/w .

Given the distance (of the leftmost point) from the right x , the needle only intersects at the right for angles between 0 to $\arccos(x/l)$. That is with probability $\arccos(x/l)/(\pi/2)$. We can ignore the negative angle case as it's symmetrical.

Compute the average probability over the region where the needle possibly intersects as:

$$1/l * \int \arccos(x/l)/(\pi/2), (x, 0, l)$$

Pulling out the constant, integrating by parts, and evaluating gives $2/\pi$.

The leftmost position and the angle are independent, so we intersect a border with probability $2l/(w \pi)$.

Dimitri's Solution

Dimitri also alerted me that [Jane Street Puzzles](#) asked a similar problem, though they only ask for an approximation instead of an exact solution. I take this to mean that I think more highly of my readers than Jane Street thinks of its readers.

My Solution I am a straightforward 'gal. As such, I solved the problem in a straightforward way. My solution has the same general form as Dimitri's, except I formulated the condition of intersection with $\cos$ instead of $\arccos$, which saved me the trouble of integration by parts. I took my two random variables to be x, θ where

$$x = \text{distance needle's center and closest line}$$

$$\theta = \text{angle of orientation}$$

Both are uniform, continuous, independent random variables, so all I had to do was integrate over the joint PDF.

My perfunctory calculation is not worth the valuable real estate that is this newsletter. Thus, I present to you a truly unbelievable proof of Buffon's Needle Problem given by Barbier in 1860. Barbier's proof is also referred to as 'Buffon's Noodle.'

Barbier's Solution (Buffon's Noodle)

Barbier's proof exploits the linearity of the expected value operator E . It begins by defining a general function $E : \mathbb{R} \rightarrow \mathbb{N}$ and variable p_i

$E(l)$ = the expected number of crossings for a needle of length l .
 p_i = probability that the needle crosses i times

It follows that

$$\begin{aligned} E &= (1)p_1 + (2)p_2 + (3)p_3 + \dots \\ &= \sum_{k=1}^{\infty} kp_k \end{aligned}$$

Since we assume $l \leq w$, a single needle can only make, at maximum, one intersection. Therefore, in our case, $E(l) = p_1$.

To be sure, there are exactly two cases where the needle can cross a line more than once: (1) The needle falls exactly on a line (2) A needle of length w falls perpendicular to the lines such that its two endpoints intersect with two lines. Since each case occurs with expected probability 0, it holds that $E(l) = p_1$.

Returning to the proof, notice that E is linear. If we split the needle of length l into two parts x and y , the expected values of each event satisfy linearity. For any needle $\lambda \in \mathbb{R}$, we have

$$\begin{aligned} E(x+y) &= E(x) + E(y) \\ E(\lambda l) &= \underbrace{E(l) + E(l) + E(l) + \dots + E(l)}_{\lambda \text{ times}} \\ &= \lambda E(l) \end{aligned}$$

Now consider continuously bending the needle and gluing its endpoints such that it is now in the shape of a circle with radius $\frac{w}{2}$. Call this circle S . This new needle is referred to as 'Buffon's Noodle.' The expected number of crossings for the circle is always 2.

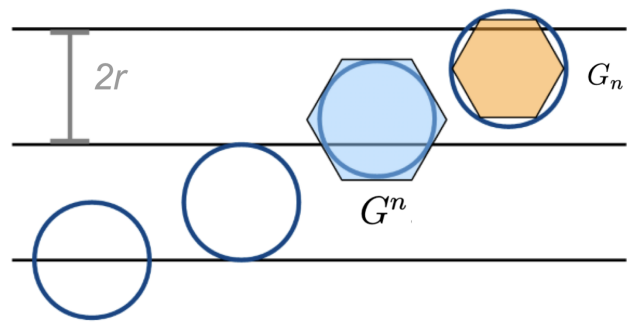
Now bound the circle from above and below with inscribed G_n and circumscribed G^n n -gons. Each n -gon is composed of n straight needles of corresponding lengths $\lambda_i \in \mathbb{R}$. Since E is linear, the expected number of crossings for the entire polygon is the sum of the expected number of crossings for each side. Thus,

$$\begin{aligned} E(G) &= \lambda_1 E(1) + \lambda_2 E(1) + \dots + \lambda_n E(1) \\ &= (\lambda_1 + \lambda_2 + \dots + \lambda_n) E(1) \end{aligned}$$

But $(\lambda_1 + \lambda_2 + \dots + \lambda_n)$ is just the *perimeter* (*per*) of the n -gon G . Thus, for any n -gon G ,

$$E(G) = \text{per}(G) \cdot E(1)$$

Return to the circle which is now bounded by an inscribed n -gon G_n and a circumscribed n -gon G^n .



It follows that

$$E(G_n) \leq E(S) \leq E(G^n)$$

$$\Rightarrow \text{per}(G_n) \cdot E(1) \leq 2 \leq \text{per}(G^n) \cdot E(1)$$

Let $n \rightarrow \infty$. The inscribed and circumscribed polygons tend closer to the circle S , where $\text{perimeter}(S) = 2\pi(\frac{w}{2}) = \pi w$. As $n \rightarrow \infty$,

$$\begin{aligned} \text{per}(G_n) \cdot E(1) &= \text{per}(S) \\ &= \text{per}(G^n) \cdot E(1). \end{aligned}$$

$$\pi w \cdot E(1) \leq 2 \leq \pi w \cdot E(1)$$

Solving for $E(1)$, we have

$$E(1) = \frac{2}{\pi w}$$

This gives the expected number of crossings for a needle of length 1. In the case where $l \leq w$, since E is linear, it follows that the expected number of crossings is

$$E(l) = lE(1) = \frac{2l}{\pi w} \quad \square$$

Puzzle #5: Pythagorean Tuning

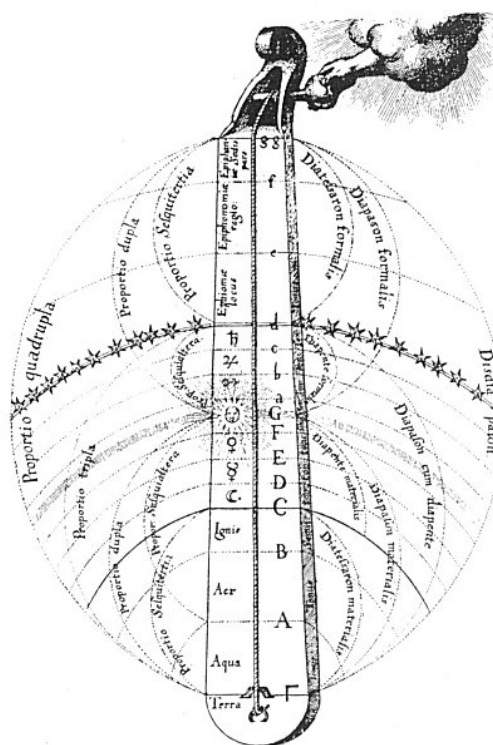
Pythagorean tuning is a system of musical tuning in which the frequency ratios of all intervals are determined by choosing a sequence of fifths which are "pure" or perfect, with ratio 3:2. The system dates back to Ancient Mesopotamia, though it has been widely mis-attributed to the Ancient Greeks by modern music theorists.

Consider the Pythagorean tuning system, where tones are generated by placing points on the unit circle at fractional positions $n\theta$ for $n = 0, 1, 2, \dots, 12$. Let $\theta = \frac{7}{12} + \delta$ be an irrational number close to $\frac{7}{12}$. This models a 'just' perfect fifth.

1. Determine how many distinct intervals (arc lengths) exist between adjacent tones in this sequence.
2. What are the possible numbers of distinct interval lengths?
3. What does this say about the structure of the tuning system?

Music held a deep cosmological significance for Pythagoras. He thought of the universe as an immense monochord, with its single string connected at its upper end to absolute spirit and at its lower end to absolute matter. Ultimately, for Pythagoras, the universe was a cord stretched between heaven and earth.

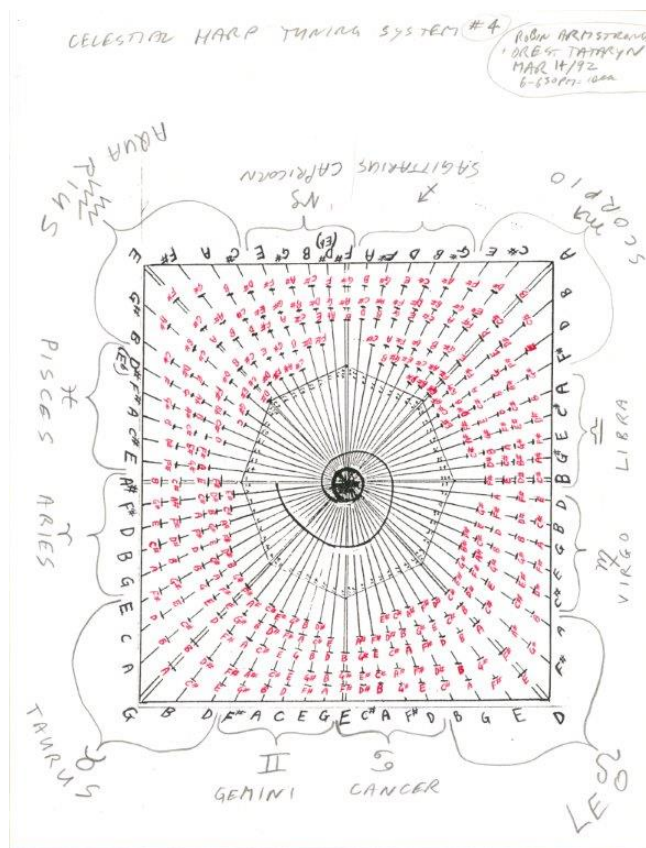
According to the concept of *the music of the spheres*, the interval between the earth and the sphere of the fixed stars was considered to be a *diapason*—the most perfect harmonic interval. Of all men, it was said that only Pythagoras heard the music of the spheres.



The Universe as Monochord

As subjects such as harmonic analysis show, music is fundamentally mathematical. Perhaps this is why many mathematicians, including my own former professors, tend to be musically inclined. Boylai played the violin. De Morgan played the flute. Grassman, a choir master, played the piano.

While mathematical ability does not guarantee musical ability, it does make intelligible the invisible movement of a harmonious tune. Though I myself am tone deaf, I seek comfort in the fact that I know Fourier analysis. Thus, the music of my ears is a little bit closer to the music of the spheres. 🎵



A Celestial Tuning System for the Harp by Orest Tataryn (1992)